

Heat-fluid-solid coupling model for turbulent forced convection within porous media

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(Received 11 February 2025; accepted 2 June 2025; published 25 June 2025)

The momentum dispersion model of flows in porous media has been validated and successfully applied by Rao and Jin [*J. Fluid Mech.* **937**, A17 (2022)]. However, the modeling of heat-fluid-solid coupling in turbulent forced convection requires further development. This research identifies two primary scaling laws for the heat transfer coefficient h between the fluid and solid: $h \sim \text{Re}_p^\beta$ with $\beta = 2$ for the Darcy regime and $\beta = 0.5$ for the Forchheimer regime, both taking into account the Prandtl number. The transition between the two scaling regimes occurs around $\text{Re}_p = 1$. To effectively connect these two regimes, a bridge function is employed. The influence of thermal dispersion on energy transport is approximated using a Laplacian term, analogous to the momentum dispersion model. A Taylor expansion is utilized to calculate the effective thermal diffusivity in relation to the local Reynolds number Re_d . The first two leading-order terms of the series are adopted. The corresponding model coefficients are determined by fitting direct numerical simulation (DNS) results. The effective thermal diffusivity approaches molecular diffusivity as porosity approaches 1, while it tends toward infinity as porosity approaches zero. The macroscopic energy equation is applied in conjunction with our previous macroscopic momentum equation and has been validated across four typical scenarios. The results indicate that the proposed coupled macroscopic models maintain a high level of accuracy across a wide range of Darcy numbers, Reynolds numbers, porosities, and Prandtl numbers. The coupled macroscopic equations, though calibrated with statistically stationary DNS data, can predict not only statistically stationary temperature distributions but also overall temperature evolutions successfully, which is addressed in this research.

DOI: [10.1103/67vg-cfqb](https://doi.org/10.1103/67vg-cfqb)

I. INTRODUCTION

Porous media are made up of a solid matrix with interconnected voids that facilitate fluid flow. When the Reynolds number Re_s , defined by the seepage velocity and pore size reaches approximately the order of 100 or higher, the flow within the pore structure transitions to a turbulent state [1,2]. The turbulence can significantly enhance heat transfer, particularly in buoyancy-driven thermal convection [3,4], as evidenced by the relationship between the Nusselt number and Rayleigh number when turbulence is fully developed within a porous media structure [5]. Consequently, the interplay between turbulence and thermal convection has garnered increasing attention [6–8].

With high turbulence intensity, turbulent forced thermal convection in porous media can markedly improve heat transfer, making it advantageous for various industrial applications. This process facilitates the mechanisms of heat charging and discharging, thereby supporting thermal energy storage systems within porous media composed of rocks or bricks [9]. As a result, forced

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thermal convection is widely utilized in solar thermal collectors and heat exchangers [10], which both consist of tubes and function analogously to porous media. Therefore, the study of turbulent forced thermal convection in porous media is essential for comprehending heat transfer dynamics in these important industrial devices.

Various methods have been employed to simulate turbulent forced thermal convection within porous media, driven by the need to enhance our understanding of natural physical phenomena and to address engineering challenges. Direct numerical simulation (DNS) is recognized as one of the most effective techniques for simulating turbulent forced thermal convection, as it captures all geometric details of the porous matrix. However, despite its high accuracy, DNS is often considered impractical due to its substantial computational and temporal demands. In engineering applications, macroscopic models are favored for studying forced thermal convection due to their lower computational costs.

A complete macroscopic model of turbulent forced thermal convection in porous media should characterize the heat exchange between fluid and solid phases. A local thermal equilibrium (LTE) model is employed to analyze efficient heat and mass transfer in turbulent flows between these two phases. This model assumes that the temperature at the fluid-solid interface can be represented by a single value. However, this simplification renders the model inadequate for comprehensive theoretical and numerical studies [11,12]. This inadequacy is particularly pronounced in scenarios involving transient heat conduction processes or when significant heat is generated during specific stages, which renders the LTE model insufficient [12,13]. Furthermore, evaluating the validity of the LTE model presents challenges, as measuring the temperature difference between the solid and fluid phases is inherently difficult. Consequently, there is a pressing need to enhance the LTE model.

An alternative to the LTE model is the local thermal nonequilibrium (LTNE) model, which has garnered significant attention in both theoretical and numerical research concerning thermal convection processes in porous media [14,15]. This model accounts for the distinct temperatures of the fluid and solid phases. Quintard [11] developed a two-energy equation model that provides a viable approach for calculating the separate temperatures of the solid and fluid phases. However, implementing this model requires determining the heat transfer coefficient between these phases [16]. Kuwahara *et al.* [17] employed a single unit cell to derive the interfacial heat transfer coefficient in the asymptotic scenario of infinite conductivity within the solid phase, thereby eliminating empirical influences. Furthermore, Nakayama *et al.* [18] formulated the macroscopic energy equations by deriving useful exact solutions for steady conduction within the solid and for thermally developing flow through a semi-infinite porous medium. Based on the double-decomposition concept, de Lemos and Rocamora [19] developed a macroscopic energy equation, which was later extended by Saito and de Lemos [20] to include an additional energy equation for the solid phase. The heat transfer coefficient at the interface between the two phases was calculated concerning laminar flow through a bed formed by square rods. Subsequently, Saito and de Lemos [21] proposed a new correlation for the interfacial heat transfer coefficient applicable for Reynolds numbers up to 10^7 . The research indicates that the scaling exponents for laminar and turbulent flow differ.

An extension for the interfacial heat transfer coefficient between fluid and solid phases is the effective thermal conductivity $\tilde{\lambda}$. Prasad *et al.* [22] conducted experiments and derived a relationship $\tilde{\lambda} = \lambda_f^\phi \lambda_s^{1-\phi}$ that depends on the conductivities of both the fluid and solid, provided that these two conductivities are not significantly different from one another. However, when the solid conductivity is much larger than the fluid conductivity, the predicted conductivity is lower than the observed value, due to the challenges associated with accurately measuring a truly stagnant thermal conductivity [23]. Buonanno and Carotenuto [24] employed volume averaging to calculate the effective conductivity, taking into account particle-to-particle contact. Imadojemu and Porter [25] as well as Tavman [26] concluded that the effective conductivity is significantly influenced by thermal diffusivity and fluid conductivity.

Lee and Yang [27] employed a second-order tensor to model the conductivity of heterogeneous anisotropic porous media. Hsu [13] proposed a closure model for transient heat conduction, while Hsiao and Advani [28] examined the effects of heat dispersion. Paek *et al.* [29] concentrated on

aluminum foam, Hu *et al.* [30] investigated unconsolidated porous media, and Carson *et al.* [31] established thermal conductivity bounds for isotropic materials. The channeling effect on effective conductivity was analyzed by Braga and de Lemos [32,33]. Yu *et al.* [34] developed two effective conductivity models for porous media composed of hollow spherical agglomerates. Samantray *et al.* [35] and Reddy and Karhikeyan [36] utilized a collocated parameter model to estimate the effective thermal conductivity of two-phase materials. Singh [37] highlighted that conductivity is influenced by the morphology, structure, and interphase interactions of the constituents. Nevertheless, the modeling of effective thermal conductivity in the presence of turbulence remains ambiguous.

This paper aims to investigate the modeling of the interfacial heat transfer coefficient and the effective thermal conductivity for turbulent forced convection within homogeneous and isotropic porous media. The macroscopic energy equation is coupled with the macroscopic momentum equation and validated against DNS results considering the detailed pore-scale geometry of the porous media. The coupled macroscopic equations are employed to predict real-time temperature changes. This research seeks to derive and validate an effective macroscopic energy equation.

This paper is organized as follows: Section II discusses the governing equations, which include both microscopic and macroscopic equations, along with the numerical methods and the determination of model coefficients. In Sec. III, the developed macroscopic models are applied to simulate turbulent flow and heat transfer in four representative scenarios, thereby demonstrating their utility. A further application of the coupled macroscopic equations is presented in Sec. IV. Finally, Sec. V offers the conclusions.

II. GOVERNING EQUATIONS, NUMERICAL METHODS, AND MODEL COEFFICIENTS

A. Governing equations for microscopic equations

The governing equations for microscopic simulations are the transient Navier-Stokes equations, as presented in Eqs. (1)–(3):

$$\frac{\partial u_i}{\partial x_i} = 0, \quad (1)$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + g_i + \nu \frac{\partial^2 u_i}{\partial x_j^2}, \quad (2)$$

$$\frac{\partial T_f}{\partial t} + u_i \frac{\partial T_f}{\partial x_i} = \dot{q} + \alpha \frac{\partial^2 T_f}{\partial x_i^2}. \quad (3)$$

It is assumed that the fluid is incompressible, and the heat source term is included. Notably, the temperature of the porous matrix is maintained as a constant value across all simulation cases. As a result, no energy equation for the solid phase is provided. Nevertheless, the effects of the solid phase will be incorporated into the heat transfer coefficient discussed in Sec. II B.

The dimensions of the variables velocity u_i , distance x_i , time t , density ρ , pressure p , applied body force g_i , kinematic viscosity of the fluid, ν , temperature T_f , heat source term $\dot{q}$, and thermal diffusivity α in Eqs. (1)–(3) are $[\text{L T}^{-1}]$, $[\text{L}]$, $[\text{T}]$, $[\text{M L}^{-3}]$, $[\text{M L}^{-1} \text{T}^{-2}]$, $[\text{L T}^{-2}]$, $[\text{L}^2 \text{T}^{-1}]$, $[\text{K}]$, $[\text{K T}^{-1}]$ and $[\text{L}^2 \text{T}^{-1}]$ respectively. Here L, T, M, and K can represent any dimensions of length, time, mass, and temperature, as they are not explicitly defined. The applied body force g_i is defined as the negative of the applied pressure gradient divided by the density. The heat source term $\dot{q}$ is defined as the rate of heat generation per unit volume divided by the density and specific heat capacity.

Given that the complexities of heat transfer within porous media primarily reside in the fluid, and considering that the temperature of the solid can be determined immediately after solving for fluid motion, this study predominantly focuses on the fluid region. To simplify the analysis, the temperature of the solid is fixed at a specific value T_s , allowing it to act as a heat sink or source

when the surrounding fluid temperature is higher or lower than the fixed value T_s . The corresponding applications can be identified in two common scenarios: (1) when the solid has a uniform initial temperature and a significantly larger heat capacity than the fluid, resulting in negligible temperature variation over a limited time frame, and (2) when the solid is interconnected with significantly higher heat conductivity than the fluid and is maintained in contact with a large thermal reservoir at a fixed temperature, allowing its surface to function as an approximately isothermal boundary. This approach is analogous to the commonly employed condition where the solid possesses a velocity of zero, thereby functioning as a momentum source.

The purpose of conducting microscopic simulations is to validate the coupled macroscopic equations. By taking into account all scales of turbulent motion down to the Kolmogorov scale, as well as the intrinsic geometry of the porous matrix, the DNS results are regarded as physically accurate.

B. Governing equations for macroscopic equations

The macroscopic equations for simulating flow in porous media can be derived through both time and volume averaging of the transient Navier-Stokes equations [38,39]. In contrast, our derivation relies solely on volume averaging Eqs. (1)–(3) utilizing the theorem of local volumetric average [2,40]. The averaging volumes are defined as representative elementary volumes (REV), which are sized appropriately to accurately capture the pore-scale properties of the entire volume under consideration. The macroscopic equations are presented below:

$$\frac{\partial(\phi\langle u_i \rangle^\gamma)}{\partial x_i} = 0, \quad (4)$$

$$\frac{\partial(\phi\langle u_i \rangle^\gamma)}{\partial t} + \frac{\partial(\phi\langle u_i \rangle^\gamma \langle u_j \rangle^\gamma)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial(\phi\langle p \rangle^\gamma)}{\partial x_i} + \phi g_i - \phi R_i + \nu \frac{\partial^2(\phi\langle u_i \rangle^\gamma)}{\partial x_j^2} - \frac{\partial(\phi\langle \tilde{u}_i \tilde{u}_j \rangle^\gamma)}{\partial x_j}, \quad (5)$$

$$\frac{\partial(\phi\langle T_f \rangle^\gamma)}{\partial t} + \frac{\partial(\phi\langle u_i \rangle^\gamma \langle T_f \rangle^\gamma)}{\partial x_i} = \phi \dot{q} - \phi H + \alpha \frac{\partial^2(\phi\langle T_f \rangle^\gamma)}{\partial x_i^2} - \frac{\partial(\phi\langle \tilde{u}_i \tilde{T}_f \rangle^\gamma)}{\partial x_i}. \quad (6)$$

Within these equations, porosity ϕ represents the fraction of void space in the porous media relative to its total volume. The total drag denoted by $R_i = \frac{1}{\phi \rho \Delta V} \int_A n_i p dS - \frac{2\nu}{\phi \Delta V} \int_A n_j \frac{\partial_i u_j + \partial_j u_i}{2} dS$ results from the effects of the porous matrix and comprises both pressure drag and viscous drag. The vector n_i indicates the normal direction from the fluid to the porous matrix at the solid-fluid interface, while A refers to the interfacial area between the solid porous matrix and the fluid. ΔV refers to the volume of one REV. The interphase energy resistance denoted by $H = -\frac{1}{\phi \Delta V} \int_A \alpha n_i \partial_i T_f dS$ is attributed to the porous matrix and encompasses heat transfer between the two phases. In this research, the detailed geometry of the porous matrix is taken into account.

$\langle \varphi \rangle^\gamma$ denotes the intrinsic average value for the fluid and φ could represent velocity and temperature. The spatial deviation $\tilde{\varphi} = \varphi - \langle \varphi \rangle^\gamma$ is defined as the difference between the microscopic value and the intrinsic volume-averaged value within the REV. The terms $\phi\langle \tilde{u}_i \tilde{u}_j \rangle^\gamma$ and $\phi\langle \tilde{u}_i \tilde{T}_f \rangle^\gamma$ represent respectively momentum dispersion and thermal dispersion, accounting for the effects of spatial deviations in the microscopic velocity and temperature fields on the macroscopic flow and heat transfer. However, the terms R_i , H , $\phi\langle \tilde{u}_i \tilde{u}_j \rangle^\gamma$, and $\phi\langle \tilde{u}_i \tilde{T}_f \rangle^\gamma$ are not inherently closed within the framework of the governing equations. To address this issue, modeling efforts are necessary to develop expressions for these terms based on the detailed geometry and properties of the porous matrix.

C. Modeling for macroscopic momentum equation

The total drag R_i is decomposed into two components: the surface integrals of the intrinsic average pressure and viscous forces R_i^γ , and the surface integrals of the spatial deviations of pressure and viscous forces $\tilde{R}_i$. R_i^γ can be approximated to zeroth order of the corresponding

values multiplied by the negative porosity gradient. Given that the porosity is considered a constant value for macroscopic simulation, R_i^γ equals zero. Consequently, the total drag $R_i = \tilde{R}_i$ and is approximated using the Darcy equation and Forchheimer equation [41,42], as presented in Eq. (7). The Darcy equation is applicable when the flow velocity is sufficiently low, while the Forchheimer equation modifies the Darcy equation to account for turbulent flow conditions. In our previous papers [2,40,43], the coefficient value c_F is approximated using a Taylor expansion, as demonstrated in Eq. (8):

$$R_i \approx \hat{R}_i = R_{Di} + R_{Fi} = \frac{\nu}{K} u_{Di} (1 + c_F \text{Re}_K), \quad (7)$$

$$c_F = c_{F1} \text{Re}_K^0 + c_{F2} \text{Re}_K^1 + c_{F3} \text{Re}_K^2 + \cdots + c_{Fn} \text{Re}_K^{n-1} + o(\text{Re}_K^n). \quad (8)$$

Within the above equations, $u_{Di} = \phi \langle u_i \rangle^\gamma$ denotes the seepage velocity and $\text{Re}_K = |\mathbf{u}_D| \sqrt{K} / \nu$. The permeability K represents the capacity of a porous medium to permit fluid to pass through it. c_F is a dimensionless coefficient, which accounts for the nonlinear increase of the form drag with the velocity.

A Laplacian term is adopted to account for the effect of the drag, $\phi(R_i - \hat{R}_i)$, molecular diffusion, and the momentum dispersion $\phi \langle \tilde{u}_i \tilde{u}_j \rangle^\gamma$, expressed as [2,40]

$$\phi(R_i - \hat{R}_i) + \nu \frac{\partial^2 (\phi \langle u_i \rangle^\gamma)}{\partial x_j^2} - \frac{\partial (\phi \langle \tilde{u}_i \tilde{u}_j \rangle^\gamma)}{\partial x_j} \approx \tilde{\nu} \frac{\partial^2 u_{Di}}{\partial x_j^2}, \quad (9)$$

$$\tilde{\nu} / \nu = c_{B1} \text{Re}_d^0 + c_{B2} \text{Re}_d^1 + \cdots + c_{Bn} \text{Re}_d^{n-1} + o(\text{Re}_d^n). \quad (10)$$

The effective viscosity $\tilde{\nu}$ is approximated by the Taylor expansion and $\text{Re}_d = K |s_{Dij}| / \nu$, where the magnitude of the strain rate $|s_{Dij}| = (2s_{Dij}s_{Dij})^{1/2}$. The effective viscosity significantly influences the velocity within the first REV near the boundary. Only the two leading terms of c_F and $\tilde{\nu}$ are adopted, and the corresponding values are presented by Rao *et al.* [2,40]

The macroscopic momentum equation has been validated by Rao *et al.* [2,40,44]. Consequently, this study directly applies the macroscopic momentum equation to simulate both laminar and turbulent flows. Furthermore, the macroscopic momentum equation is coupled with the macroscopic energy equation to simulate temperature distribution and predict the time evolution of overall temperature. The following section discusses the derivation of the macroscopic energy equation.

D. Modeling for macroscopic energy equation

This section aims to model the interphase energy resistance H and thermal dispersion $\phi \langle \tilde{u}_i \tilde{T}_f \rangle^\gamma$ to close the macroscopic energy equation for porous media.

1. Modeling for interphase energy resistance

Typically, the interphase energy resistance H is modeled using the effective interfacial convective heat transfer coefficient h , which considers the heat transfer $\frac{1}{\phi \Delta V} \int_A \alpha n_i \partial_i T_f dS$ between the fluid and solid phases. Consequently, the model is presented below:

$$H = -\frac{1}{\phi \Delta V} \int_A \alpha n_i \partial_i T_f dS \approx \hat{H} = ha(\langle T_f \rangle^\gamma - \langle T_s \rangle^\gamma), \quad (11)$$

where $a = A / \Delta V$ represents the ratio of the surface area of the porous matrix to the volume within one REV.

To determine the heat transfer coefficient h , Kuwahara *et al.* [17] solved the governing equations for a regular porous medium by considering an infinite array of solid square rods. Periodic boundary conditions were applied to obtain the temperature distribution under fully developed flow conditions. The numerical correlation for the interfacial convective heat transfer coefficient is presented

below:

$$\frac{hD}{\alpha} = \left(1 + \frac{4(1-\phi)}{\phi}\right) + \frac{1}{2}(1-\phi)^{1/2}\text{Re}^{0.6}\text{Pr}^{1/3}. \quad (12)$$

Wakao *et al.* [45] conducted a series of experiments and found a consistent scaling law relating to the variables h and Re . Saito and de Lemos [20] derived the interfacial heat transfer coefficient for laminar flow in the staggered array of square rods. Subsequently, Žukauskas [46] proposed a correlation applicable within a specified range $2 \times 10^5 < \text{Re} < 2 \times 10^6$ for turbulent flow, as illustrated below:

$$\frac{hD}{\alpha} = 0.022\text{Re}^{0.84}\text{Pr}^{0.36}. \quad (13)$$

The aforementioned correlation $h \sim \text{Re}^\beta$ indicates a scaling exponent β associated with either laminar or turbulent flow. This exponent β varies with changes in Reynolds numbers, which limits its applicability. Currently, no model for h can accurately represent both laminar and turbulent flow simultaneously.

In this research, we identify two distinct scaling exponents: one, $\beta = 2$, applicable for the Darcy regime (low Reynolds number), and another, $\beta = 0.5$, for the Forchheimer regime (moderate to large Reynolds numbers). To seamlessly integrate these two scaling exponents, we propose a bridge function, as demonstrated below:

$$\frac{(h - h_0)\sqrt{K}}{\alpha} = \frac{c_{h1}\text{Re}_p^{1/2}}{1 + (\text{Re}_p/c_{h2})^{-3/2}}, \quad \frac{h_0\sqrt{K}}{\alpha} = c_{h0}. \quad (14)$$

Here, $\text{Re}_p = \text{Re}_K \text{Pr}$, where $\text{Pr} = \nu/\alpha$ is the Prandtl number defined as the ratio of molecular viscosity to molecular thermal diffusivity. In this research, the molecular viscosity is held constant; therefore, as the Prandtl number increases, the molecular thermal diffusivity decreases, leading to a higher temperature distribution. Based on our previous research [44], when porosity is within the low to medium range, turbulence is constrained by the pore size s . Consequently, the characteristic length is identified as the pore size s . In this study, we utilize the characteristic length $\sqrt{K}$ to describe the pore size, as it demonstrates a linear relationship with the pore size s when the shape of the porous matrix remains constant, similar to modeling of the total drag R_i . Additionally, it is easier to determine for a porous matrix. The coefficients c_{h0} , c_{h1} , and c_{h2} are determined by fitting the DNS results, while h_0 represents the heat transfer coefficient in the absence of thermal convection. Therefore, the interphase energy resistance $\hat{H}$ is modeled as shown in Eq. (15), where $T_D = \phi\langle T_f \rangle^\gamma$, similar to the model of total drag $\hat{R}$ in Eq. (7):

$$\hat{H} = \frac{\alpha}{K_T} T_D (1 + c'_{h1} \text{Re}_p^{1/2} (1 + (\text{Re}_p/c_{h2})^{-3/2})^{-1}), \quad K_T = \frac{\alpha\phi}{h_0 a} = \frac{\phi\sqrt{K}}{c_{h0} a}, \quad c'_{h1} = \frac{c_{h1}}{c_{h0}}. \quad (15)$$

2. Modeling for the thermal dispersion

The gradient of macroscopic temperature influences the modelling error of energy resistance $\phi(H - \hat{H})$, molecular thermal diffusion $\alpha\partial\phi\langle T_f \rangle^\gamma/\partial x_i$, and thermal dispersion $\phi\langle \tilde{u}_i \tilde{T}_f \rangle^\gamma$. A first-order tensor $\tilde{\alpha}\partial T_D/\partial x_i$ is introduced to account for these effects, where $\partial T_D/\partial x_i$ is gradient of the macroscopic temperature as shown below:

$$\phi(H - \hat{H}) + \alpha \frac{\partial^2 \phi\langle T_f \rangle^\gamma}{\partial x_i^2} - \frac{\partial(\phi\langle \tilde{u}_i \tilde{T}_f \rangle^\gamma)}{\partial x_i} = \tilde{\alpha} \frac{\partial^2 T_D}{\partial x_i^2}, \quad (16)$$

$$\tilde{\alpha}/\alpha = c_{T1}\text{Re}_d^0 + c_{T2}\text{Re}_d^1 + c_{T3}\text{Re}_d^3 + \dots + c_{Tn}\text{Re}_d^{n-1} + o(\text{Re}_d^n). \quad (17)$$

Similar to the modeling of momentum dispersion, the effective thermal diffusivity $\tilde{\alpha}$ is approximated using a Taylor expansion with respect to the local Reynolds number, $\text{Re}_d = K|s_{Di,j}|/\nu$. $\sqrt{K}$ is identified as the characteristic length and $\sqrt{K}|s_{Di,j}|$ is utilized as the characteristic velocity. This

approach is analogous to the mixing length model of Prandtl [47]. $c_{T1}, c_{T2}, c_{T3}, \dots, c_{Tn}$ represent the Taylor coefficients. In this study, we only adopt the two leading-order terms as shown in Eq. (18). The normalized value $\tilde{\alpha}/\alpha$ approaches a constant value c_{T1} as $\text{Re}_d \rightarrow 0$ and increases linearly with increasing Re_d :

$$\tilde{\alpha}/\alpha = c_{T1}\text{Re}_d^0 + c_{T2}\text{Re}_d^1. \quad (18)$$

Tortuosity has been introduced to account for variations in pore-scale geometries, but its determination remains challenging. Furthermore, no clear relationship has been established between the model coefficients and tortuosity [40]. Additionally, this study only considers a porous matrix composed of cubes. Consequently, this study does not account for tortuosity.

In summary, the macroscopic equations are presented below:

$$\frac{\partial u_{Di}}{\partial x_i} = 0, \quad (19)$$

$$\frac{\partial u_{Di}}{\partial t} + \frac{\partial(u_{Di}u_{Dj}/\phi)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial(\phi\langle p \rangle^\gamma)}{\partial x_i} + \phi g_i - \phi \hat{R}_i + \tilde{v} \frac{\partial^2 u_{Di}}{\partial x_j^2}, \quad (20)$$

$$\frac{\partial T_D}{\partial t} + \frac{\partial(u_{Di}T_D/\phi)}{\partial x_i} = \phi \dot{q} - \phi \hat{H} + \tilde{\alpha} \frac{\partial^2 T_D}{\partial x_i^2}. \quad (21)$$

The corresponding coefficients for $\hat{R}_i$ and $\tilde{v}$ were established in previous research [2,40], while the coefficients for $\hat{H}$ and $\tilde{\alpha}$ will be discussed and determined in the following section.

E. Numerical methods

In this part, two methods are applied in the simulations: (1) the finite volume method (FVM), to solve microscopic (DNS) and macroscopic equations, and (2) the spectral method (SM), to solve macroscopic equations.

The finite volume method is utilized in this paper to solve both microscopic (DNS) and macroscopic equations. The solver pimpleFoam, based on OpenFOAM 18.12, has been modified to couple flow and temperature. A second-order central difference scheme is employed for spatial discretization of velocity and temperature, achieved through the interpolation of variables at the interfaces of grid cells. The Poisson equation is solved at each time step to update the pressure, while the Pressure-Implicit with Splitting of Operators (PISO) scheme ensures accurate coupling between pressure and velocity.

The spectral method is used to solve the macroscopic equations via a solver developed in matlab. This method represents the solution as a Fourier series of sinusoids, with coefficients selected to best satisfy the differential equation. Compared to the finite difference method, spectral methods are more efficient and easier to implement, particularly in cases requiring high accuracy for smooth solutions in simple computational domains.

The FVM and SM solvers have been previously validated and verified in earlier studies [2,40,44]; thus, they are directly employed in this work. DNS solutions will be utilized to determine the parameters and validate the results of the macroscopic model. Furthermore, typical DNS cases are assessed to quantify the uncertainty arising from mesh dependence. These details will be elaborated upon in the following section.

F. Coefficients determination for macroscopic energy equation

The corresponding coefficients in the macroscopic momentum equation have been established in a previous papers [2,40]. Therefore, this study focuses on determining the model coefficients for the heat transfer coefficient h and effective thermal diffusivity $\tilde{\alpha}$ in the macroscopic energy equation. To ascertain the model coefficients, we employ DNS to compute turbulent forced thermal convection within the porous media and the model coefficients are determined by fitting DNS results.

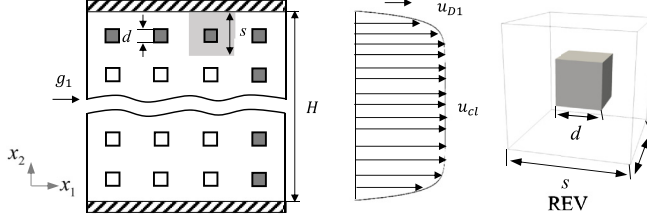


FIG. 1. The schematic of the porous matrix made up of cubes and a representative elementary volume (REV) for the small porous element.

This study employs a porous matrix composed of arrays of cubes, as illustrated in Fig. 1. The porous matrix consists of 128 REVs arranged in a configuration $4s \times 8s \times 4s$, where the pore size s is defined by the distance between the centers of two adjacent solid elements. Periodic boundary conditions are applied in both the streamwise (x_1) and spanwise (x_3) directions. The number of REV in the wall-normal direction is sufficiently large to ensure that the flow in the central region remains unaffected by the walls. The molecular viscosity ν is set to $0.002 \text{ L}^2 \text{ T}^{-1}$. The parameter g_1 is varied to achieve different Reynolds numbers, while the heat source term $\dot{q}$ is maintained at a constant value of 1 K T^{-1} . The temperature of both the wall boundary and the solid porous matrix is set to zero across all cases. The simulations are conducted using the FVM, with grid points uniformly distributed throughout the domain. Details regarding the verification of mesh independence are provided in the Appendix.

Figure 2 illustrates the instantaneous velocity magnitude, temperature, and REV-averaged temperature $\langle T_f \rangle^{V, x_1, x_3}$ where $\langle \dots \rangle^{V, x_1, x_3}$ refers to the time-averaged, REV volume-averaged, streamwise-averaged (x_1), and spanwise-averaged (x_3) results for the fully developed turbulent flow. It is evident that the flow is fully developed and transitions into turbulence. Figure 2(c) indicates that the effective thermal diffusivity significantly influences the temperature distribution in the first REV adjacent to the wall, which aligns with the pore-scale prevalence hypothesis (PSPH) [48]. Consequently, for the fully developed turbulence, the macroscopic energy equation in the central region of the channel can be simplified as

$$0 = \phi \dot{q} - \phi h a \langle T_f \rangle^\gamma. \quad (22)$$

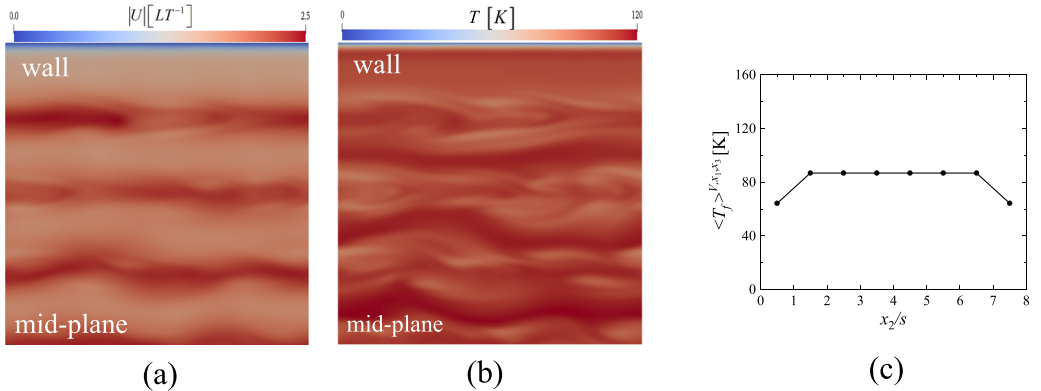


FIG. 2. (a) Instantaneous velocity magnitude, (b) instantaneous temperature, and (c) the REV-averaged temperature $\langle T_f \rangle^{V, x_1, x_3}$ versus the normalized distance from the wall. $\text{Re}_{cl} = 5736$, $\phi = 0.76$, $\text{Pr} = 5$, turbulent flow.

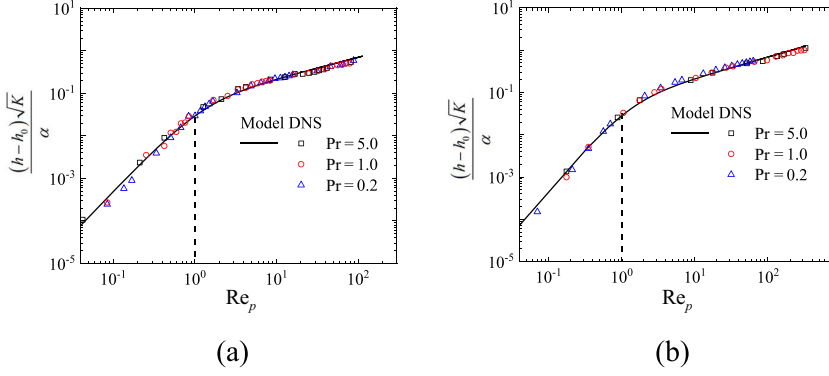


FIG. 3. The determination of the heat transfer coefficient h by fitting DNS results including $Pr = 5, 1$, and 0.2 . (a) $\phi = 0.64$, $c_{h0} = 1.88$, $c_{h1} = 0.07$, $c_{h2} = 1.26$; (b) $\phi = 0.76$, $c_{h0} = 1.89$, $c_{h1} = 0.069$, $c_{h2} = 1.28$.

To determine the value of h , we first specify g_1 and subsequently calculate the macroscopic $\phi\langle T_f \rangle^\gamma$ using DNS results. We then use the equation $h = \dot{q}/a\langle T_f \rangle^\gamma$ to obtain the value of h . A series of DNS cases with $Pr = 5, 1$, and 0.2 are conducted, then the coefficients c_{h1} and c_{h2} of the model in Eq. (14) are obtained directly by fitting DNS results, as shown in Fig. 3. It is important to note that the obtained values of the coefficients c_{h1} and c_{h2} correspond specifically to the respective porosity and are not necessarily universal across all porosities. However, fitting results show that the scaling transition parameter c_{h2} is almost the same across different porosities and Prandtl numbers, indicating the appropriateness of using Re_p for the modeling of H .

In the first REV close to the wall, the macroscopic energy equation is simplified as

$$0 = \phi\dot{q} - \phi h a \langle T_f \rangle^\gamma + \tilde{\alpha} \frac{\partial^2 \phi \langle T_f \rangle^\gamma}{\partial x_i^2}. \quad (23)$$

The values of c_{T1} and c_{T2} in Eq. (18) are obtained by fitting DNS results. The error of h within the central region may affect the accuracy of c_{T1} and c_{T2} . To address this uncertainty, the values of c_{h1} and c_{h2} are modified to ensure that the calculated temperature T_{fcl} at the centerline aligns with the DNS result of the corresponding single case.

Initially, we set the velocity to zero, which results in an initial condition $Re_d = 0$. The DNS results obtained without thermal convection are utilized to determine c_{T1} , which varies with porosity as presented in Fig. 4(a). The function of c_{T1} can be reasonably approximated by Eq. (24), which

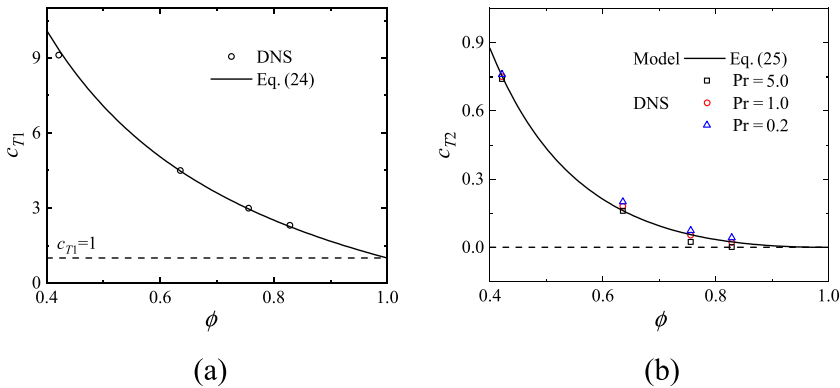


FIG. 4. The empirical equations versus porosity: (a) c_{T1} ; (b) c_{T2} .

indicates that c_{T1} is independent of the Prandtl number. It is observed that c_{T1} approaches 1 as the porosity ϕ increases asymptotically to 1 (consistent with the microscopic equation, Eq. (3)), while it tends toward infinity as ϕ decreases asymptotically to zero, aligning with physical expectations.

The coefficient c_{T2} is determined by fitting the DNS results for $\text{Re}_p \gg 1$. Theoretically, the coefficient c_{T2} should be related with Prandtl number. However, when the porosity is not large, the interphase energy resistance $\hat{H}$ still constitutes a substantial portion in the first REV, similar to total drag $\hat{R}$ [2]. Within the examined range of the Prandtl number, its effect on the coefficient c_{T2} is negligible. Therefore, for porous matrices composed of cubes, c_{T2} can be approximated using Eq. (25). Notably, c_{T2} approaches zero as the porosity ϕ increases asymptotically to 1, while it approaches infinity as ϕ decreases asymptotically to zero, as shown in Fig. 4(b):

$$c_{T1} = 6.07 \frac{1 - \phi}{\phi} + 1, \quad (24)$$

$$c_{T2} = 0.61 \frac{(1 - \phi)^2}{\phi^{3/2}}. \quad (25)$$

III. VALIDATION AND DISCUSSION

This section applies DNS to validate the macroscopic energy equation and analyzes the results. The developed macroscopic energy equation is coupled with a macroscopic momentum equation and is used to simulate turbulent forced thermal convection within a channel filled with isotropic and homogeneous porous media characterized by one or two length scales. The results and relevant analyses are presented below.

A. Varying Darcy numbers

This section aims to demonstrate the applicability of the thermal dispersion model across a broad range of Darcy number, Da . In the simulation, we utilize the schematic of the porous medium presented in Fig. 1, along with the DNS results of REV-averaged temperature $\langle T_f \rangle^V$ as illustrated in Fig. 2. The Darcy number is defined as follows:

$$\text{Da} = \frac{K}{H^2}. \quad (26)$$

The centerline Reynolds number Re_{cl} is based on the channel height H and the centerline macroscopic velocity in the channel, u_{cl} , and is defined as follows:

$$\text{Re}_{cl} = \frac{u_{cl}H}{\nu}. \quad (27)$$

We fixed the centerline Reynolds number Re_{cl} and reduced the pore size s , resulting in three different scale ratios H/s : 10, 20, and 40. The corresponding Darcy numbers are 1.3×10^{-4} , 3.2×10^{-5} , and 8.1×10^{-6} , respectively. Figure 5(a) illustrates that the distribution of T_D is steeper near the wall when Da is smaller. The results from the model T_D and DNS T_f are averaged in each REV, allowing for quantitative comparison. The symbol $\langle \cdot \cdot \cdot \rangle^V$ in Fig. 5 denotes the REV-averaged temperature obtained from both DNS and model results. It is evident that the results from the model are in good agreement with those obtained from DNS. The numerical results presented in Fig. 5(a) are shown in Fig. 5(b) again, but the distance from the wall, x_2 , is normalized with the pore size s instead of H . The profiles of T_D collapse into a single curve, suggesting that the characteristic length of the flow is s , which is consistent with the principles of the PSPH [48].

Figure 5(b) also illustrates the significance of the coefficient c_{T2} from the thermal dispersion model. It is important to note that neglecting c_{T2} can result in an overestimation of temperature value near the wall by approximately 6.4%. Furthermore, in the absence of the value c_{T2} , the discrepancy between the model and DNS results tends to increase with the Reynolds number Re_s ,

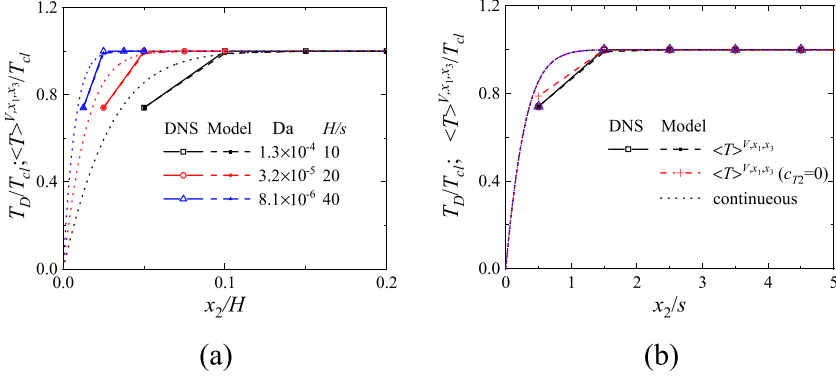


FIG. 5. The porous matrix is composed of cubes, $\phi = 0.76$, $Re_{cl} = 5736$, $Pr = 5$. Dotted lines, T_D/T_{cl} obtained from continuous model results; open symbols, $\langle T \rangle^V/T_{cl}$ from REV-averaged DNS results; solid symbols, $\langle T_D \rangle^V/T_{cl}$ from REV-averaged model results. (a) T_D/T_{cl} and $\langle T \rangle^V/T_{cl}$ versus the normalized distance x_2/H from the wall; (b) T_D/T_{cl} and $\langle T \rangle^V/T_{cl}$ versus the normalized distance x_2/s from the wall.

particularly under turbulent flow conditions. These findings highlight the necessity of incorporating the coefficient c_{T2} , especially in scenarios involving turbulence.

B. Varying Reynolds numbers, porosities, and Prandtl numbers

This section aims to further validate the macroscopic energy equation across a range of Reynolds numbers, porosities, and Prandtl numbers. The macroscopic energy equation is utilized to simulate flow within a channel filled with isotropic and homogeneous porous media, as illustrated in Fig. 1. Simulations are conducted for both laminar and turbulent flow regimes. In this section, the Reynolds number is defined based on the pore size s as regions with high porosity demonstrate increased velocity, which is calculated below:

$$Re_s = \frac{|\mathbf{u}_D|s}{\nu} \quad (28)$$

where $\mathbf{u}_D$ represents the macroscopic velocity far from the wall boundary. Figure 5 illustrates that the effective thermal diffusivity significantly influences the temperature distribution within the first REV adjacent to the wall boundary under investigation. Consequently, the macroscopic energy equation results are compared with DNS results for the volume-averaged value $\langle T \rangle^{V,x_1,x_3}$ in the first REV next to the wall, as shown in Fig. 6. In our previous studies [2,40], the volume-averaged value $\langle u_{D1} \rangle^{V,x_1,x_3}$ has already been compared. Therefore, this paper does not present a comparison of the volume-averaged velocity between the DNS and model results.

Figure 6 presents that, as the Reynolds number increases, the volume-averaged temperature decreases. This reduction is particularly evident in the lower range of Reynolds numbers, as demonstrated in Fig. 6(d) for $Pr = 5$. In contrast, within the moderate to high range of Reynolds numbers, the volume-averaged temperature shows a steady decline, as the turbulent flow becomes fully developed and stabilized in this range. Additional simulations were performed with varying porosities and Prandtl numbers. As porosity increases, effective thermal diffusivity decreases to a constant molecular thermal diffusivity, which leads to a higher temperature distribution in forced thermal convection. Although a minor disparity exists between the DNS and model results as porosity increases, the model results remain consistent with the DNS results. Furthermore, with an increasing Prandtl number at fixed molecular viscosity, molecular thermal diffusivity decreases, leading to a higher temperature distribution. And the larger discrepancy in volume-averaged temperature between the model and DNS results can be attributed to the error in c_{T2} . Nevertheless, it is

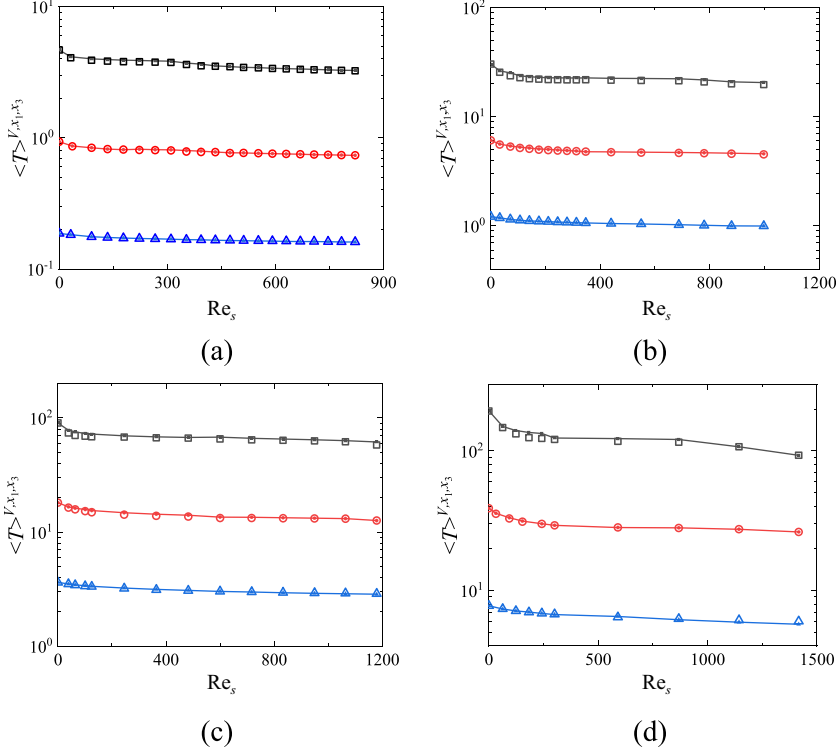


FIG. 6. $\langle T \rangle^{V, x_1, x_3}$ in the first REV next to the wall versus the Reynolds number Re_s . Open symbols represent DNS results, while solid symbols indicate macroscopic simulation results. The three lines are represented by squares (black, $Pr = 5$), circles (red, $Pr = 1$), and triangles (blue, $Pr = 0.2$). (a) $\phi = 0.42$; (b) $\phi = 0.64$; (c) $\phi = 0.76$; (d) $\phi = 0.83$.

important to emphasize that the agreement between the model and DNS results remains robust with respect to Reynolds number, porosity, and Prandtl number.

C. Flow in the porous media with two length scales

This section aims to validate the coupled macroscopic equations in more complex geometries, as illustrated in Fig. 7. In the DNS setups, the porous matrices consist of uniformly aligned cubes and bars along the x_3 direction with infinite length. It is more straightforward to recognize the role of the bars as porous matrices by also considering the channel walls as solid matrices and periodically extending the solid objects in the x_2 direction. Conversely, in the macroscopic simulation, the porous matrices are comprised solely of bars. The entire simulation domain encompasses REV configurations as $32s \times 32s \times 16s$ for the cubes. Previous research [44] indicates that no macroscopic turbulence occurs if the porosity is less than or equal to 0.93. Consequently, the computational domain in this study is sufficiently large to allow for the full development of turbulent flow. In Figs. 7(a) and 7(b), the bars have side length $d_l = 16s$, and periodic boundary conditions are applied in the streamwise (x_1) and spanwise (x_3) directions, while a wall boundary condition is implemented in two planes perpendicular to the x_2 (wall-normal) direction. In Fig. 7(c), the bars have side length $d_l/2$, and periodic boundaries are enforced in all directions. The applied body force is imposed in the x_1 direction. The overall Reynolds numbers for the porous media with two length scales are defined as $Re_l = u_m d_l / \nu$, where u_m represents the mean seepage velocity over the whole computational domain. Here, ϕ denotes the porosity of the porous matrix only composed of cubes. The heat source

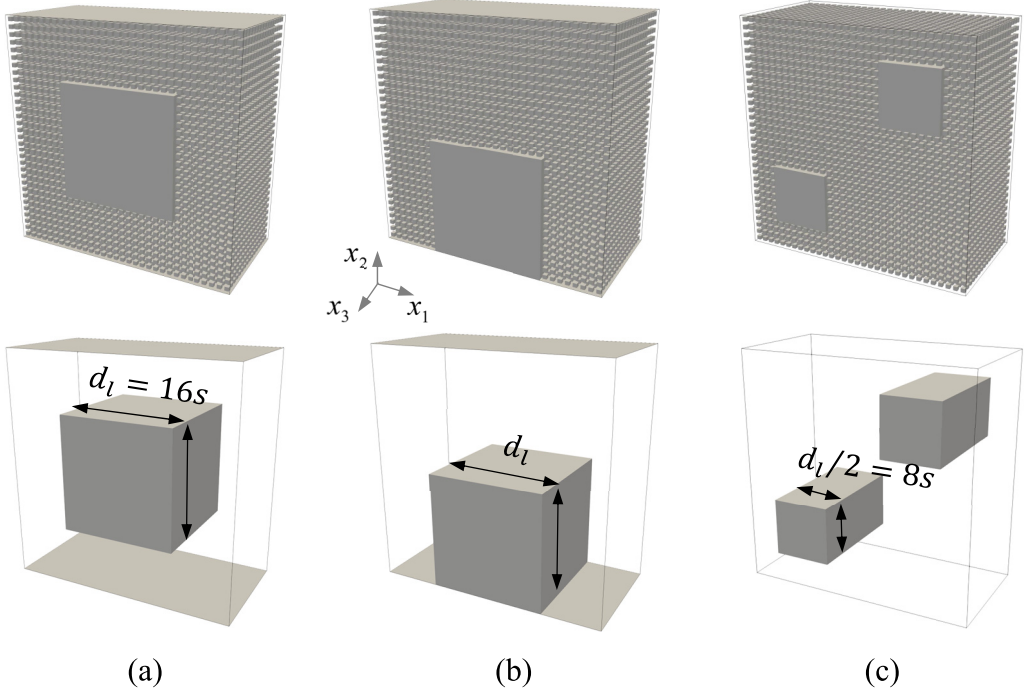


FIG. 7. Porous Matrices. Top: A porous matrix featuring two length scales (composed of cubes and bars) utilized in the DNS study. Bottom: A porous matrix with only a large pore scale (constructed from bars) for the macroscopic study. Periodic boundary conditions are applied in both the streamwise and spanwise directions. (a) The large porous element is positioned in the center of the channel, with wall boundaries at the top and bottom. (b) The large porous element is placed against one wall. (c) Two large porous elements are arranged in staggered corners, with periodic boundaries enforced in all directions.

term $\dot{q}$ is maintained at a constant value of 1 K T^{-1} , while the temperature of both the wall boundary and the solid porous matrix is set to zero.

Figure 8 illustrates the instantaneous velocity magnitude and temperature for both DNS and model results with $\text{Pr} = 1$. Corresponding to the geometries of the porous media shown in Fig. 7, the overall values of Reynolds number Re_l are 1.02×10^4 , 1.02×10^4 , and 1.18×10^4 , respectively,

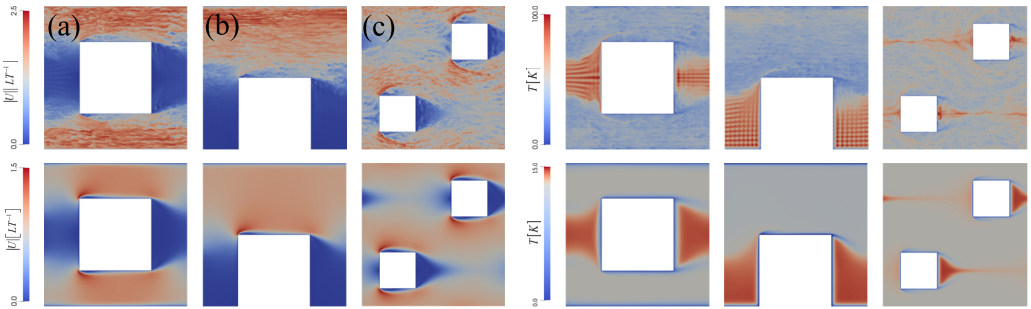


FIG. 8. Instantaneous velocity magnitude and temperature for porous medium with two length scales at $\text{Pr} = 1$. Top: DNS results; bottom: model results. (a) $\text{Re}_l = 1.02 \times 10^4$; (b) $\text{Re}_l = 1.02 \times 10^4$; (c) $\text{Re}_l = 1.18 \times 10^4$.

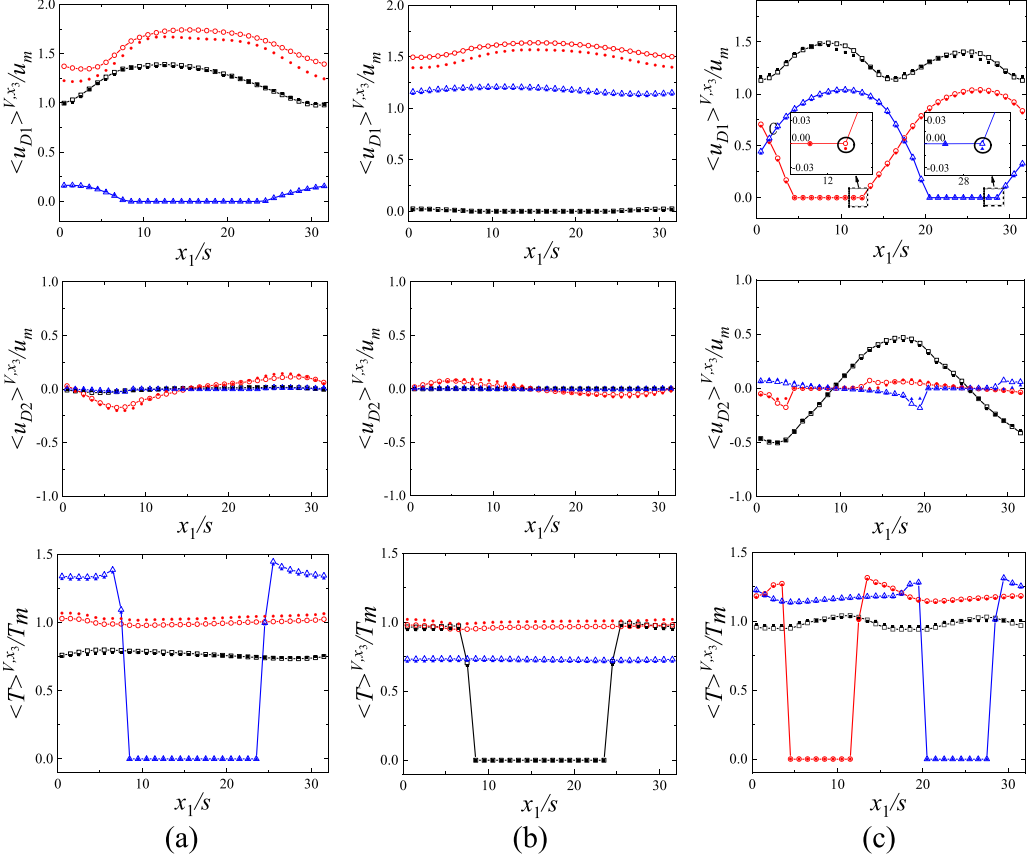


FIG. 9. Distribution of $\langle u_{D1} \rangle^{V,x_3}$, $\langle u_{D2} \rangle^{V,x_3}$, and $\langle T \rangle^{V,x_3}$ along the streamwise (x_1) lines; $\langle \dots \rangle^{V,x_3}$ refers to the time-averaged, REV volume-averaged, and spanwise-averaged results for the fully developed turbulent flow. Open symbols represent DNS results, while solid symbols indicate macroscopic simulation results, $\text{Pr} = 1$. The three lines are represented by squares (black), circles (red), and triangles (blue) in the following sequence: (a) $x_2/s = 0.5, 3.5$, and 15.5 ; (b) $x_2/s = 0.5, 24.5$, and 31.5 ; (c) $x_2/s = 0.5, 7.5$, and 23.5 .

which is sufficiently high to ensure that the turbulent flow is fully developed. These preliminary visualizations demonstrate surprising similarity of velocity and temperature distributions between DNS and model results. In the present setups, no large-scale turbulence is observed in both DNS results and macroscopic model results. This observation is reasonable, as the turbulence confined to pore size has already been modeled based on PSPH [2,40], which aligns with previous research [44].

Figure 9 illustrates the time-, REV-, and spanwise (x_3)-averaged components $\langle u_{D1} \rangle^{V,x_3}$, $\langle u_{D2} \rangle^{V,x_3}$, and $\langle T \rangle^{V,x_3}$ along the streamwise (x_1) lines. The three selected streamwise lines of each case represent those closest to the walls, farthest from the walls, or crossing the centers of bars, providing a sufficient basis for comparing DNS and model results. These quantitative results indicate satisfactory accuracy of our macroscopic model. The comparison also reveals that the differences at the line farthest from the walls are more pronounced, which may be attributed to the greater influence of the Darcy-Forchheimer term at distances farther from the wall boundary. Only the two leading terms of the Darcy-Forchheimer term in Eq. (8) are considered, resulting in overestimated drag coefficients R_1/u_{D1} and consequently smaller streamwise velocities in the model results. Consequently, this leads to a higher temperature distribution, particularly at high

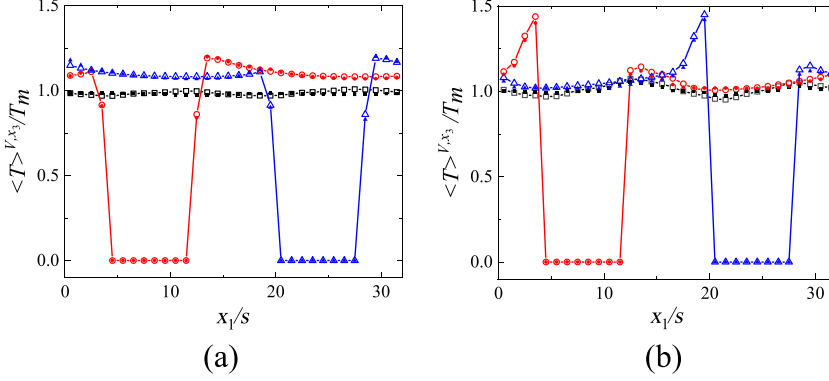


FIG. 10. Distribution of $\langle T \rangle^{V,x_3}$ from DNS and model results. Open symbols represent DNS results, while solid symbols indicate macroscopic simulation results. The three lines are represented by squares (black), circles (red), and triangles (blue) in the following sequence: $x_2/s = 0.5, 7.5$, and 23.5 . (a) $Pr = 0.2$; (b) $Pr = 5$.

Reynolds numbers, where the discrepancies between the DNS and model results become more significant. In short, the discrepancies in the REV-averaged temperature can be partly explained with the discrepancies in velocities, which might be further reduced by adding higher-order terms in the Darcy-Forchheimer term.

In contrast, the model results for the line closest to the wall align more closely with the DNS results. Given that effective thermal diffusivity has a more pronounced effect on the first REV near the wall, and considering that the modeling of thermal dispersion is a critical aspect of this study, it suggests that the thermal dispersion model is well validated.

Additionally, our findings indicate that for laminar flow, no reverse flow is observed in the wake of the porous matrix in the macroscopic simulations. In contrast, for fully developed turbulent flow, a weak reverse flow is evident in the wake of the porous matrix in the model results, as illustrated in Fig. 9(c). However, DNS results show no significant reverse flow, which can be attributed to the strong dispersion effects caused by the small porous matrix. The discrepancy between the DNS and model results may also arise from the Darcy-Forchheimer approach being less accurate than the actual conditions, as it lacks additional terms in the constitutive formulation.

D. Statistical results of $Pr = 0.2$ and 5 in porous media with two length scales

The previous sections addressed turbulent forced thermal convection in porous media with a fixed Prandtl number ($Pr = 1$). This section shifts focus to vary Prandtl numbers. A higher Prandtl number corresponds to lower molecular thermal diffusivity and a higher volume-averaged temperature within the REV when the molecular viscosity is fixed. Here, we emphasize thermal convection in conjunction with turbulent flow, which allows for a more coherent analysis. For this investigation, the porosity x_3 of the cubes is fixed at 0.83 , as this value enhances the comparison between the model results and the DNS results, thereby validating the macroscopic equations more effectively than would be possible with a lower porosity. The porous medium characterized by two distinct length scales is applied, as illustrated in Fig. 7(c), while the overall Reynolds number Re_l and other initial conditions remain the same. Figure 10, comparing the $\langle T \rangle^{V,x_3}$ distributions of DNS and model results, further indicates satisfactory estimations of temperature distribution at $Pr < 1$ and $Pr > 1$ by our macroscopic model. The discrepancy between DNS and model results at $Pr = 5$ is relatively more pronounced, indicating greater sensitivity to the velocity disparity, particularly for results obtained from the line farthest from the wall. It appears that, alongside the higher Prandtl number, the error tends to be larger. However, the maximum error is approximately 2%, which remains within acceptable limits.

IV. APPLICATION OF THE COUPLED MACROSCOPIC EQUATIONS: TRANSIENT PREDICTION OF OVERALL TEMPERATURE

This section focuses on applying the coupled macroscopic equations to predict real-time temperature changes, starting from an initial state of zero and transitioning from unsteady to statistically steady flow. The coupled macroscopic equations are not only valuable for predicting temperature distribution but also for predicting real-time changes in temperature. This capability is crucial for effective thermal management, which necessitates continuous prediction to prevent excessive and uncontrolled temperature increases, thereby ensuring the longevity and safety of related devices. However, a significant challenge remains: the lack of precise temperature prediction and the time-consuming nature of existing methods. The coupled macroscopic equations offer a promising solution by effectively balancing precision and efficiency. The following discussion highlights the utility of coupled macroscopic equations in real-time temperature prediction, particularly in scenarios characterized by high thermal conductivity, such as in liquid cooling applications.

This part will show how to apply the coupled macroscopic equations to predict real-time temperature changes. The schematic geometry of porous media with two length scales in Fig. 7(c) is applied. The schematic geometry is regarded as a complete system, and then the coupled macroscopic equations are integrated over the entire volume, as shown below:

$$\int \frac{\partial(u_{Di})}{\partial t} dV + \int \frac{\partial(u_{Di}u_{Di}/\phi)}{\partial x_j} dV = \int -\frac{1}{\rho} \frac{\partial(\phi \langle p \rangle^\gamma)}{\partial x_i} dV + \int \phi g_i dV - \int \phi \hat{R}_i dV + \int \tilde{v} \frac{\partial^2 u_{Di}}{\partial x_j^2} dV, \quad (29)$$

$$\int \frac{\partial T_D}{\partial t} dV + \int \frac{\partial(u_{Di}T_D/\phi)}{\partial x_i} dV = \int \phi \dot{q} dV - \int \phi \hat{H} dV + \int \tilde{\alpha} \frac{\partial^2 T_D}{\partial x_i^2} dV. \quad (30)$$

Since this is considered as a complete system and no macroscopic turbulence occurs based on PSPH [44,48], the macroscopic equations can be simplified into ordinary differential equations by canceling the volume V , as shown below:

$$\frac{d\bar{u}_i}{dt} = \phi g_i - \phi \hat{R}_{is}, \quad (31)$$

$$\frac{d\bar{T}}{dt} = \phi \dot{q} - \phi \hat{H}_s. \quad (32)$$

Here, $\bar{\varphi}$ denotes the overall volume average of any field φ . Among the aforementioned equations, the effective overall drag $\hat{R}_{is}$ and interphase energy resistance $\hat{H}_s$ can be modeled following the same framework introduced in our previous and present works as shown in Eqs. (7), (8), (14), and (15). The most important difference for specific implementation is that we now have the fully closed macroscopic equations [Eqs. (19)–(21)] from which $\hat{R}_{is}$ and $\hat{H}_s$ can be calculated. Therefore, instead of the highly demanding DNS, the model coefficients can now be calculated with simulation results of the coupled macroscopic equations, which can significantly reduce computational costs. The model coefficients of $\hat{R}_{is}$ can be determined using the same methodology as outlined in previous research [2,40], and those of $\hat{H}_s$ can be established as described in Sec. II F. Since the macroscopic momentum equation is coupled with the macroscopic energy equation, the overall interphase energy resistance is not constant and varies with the increasing Reynolds number Re_{ps} , indicating that these equations are interdependent. It is subsequently employed to predict the overall volume-temperature evolution. This section primarily utilizes the macroscopic energy equation [Eq. (32)] to track real-time temperature changes. Therefore, the model coefficients of $\hat{R}_{is}$ are adjusted to ensure that the macroscopic velocity aligns with the DNS velocity. The evolution of volume-averaged velocity and temperature is solved using the Dormand-Prince algorithm implanted as ode45 in MATLAB. The transient DNS is performed to validate the predicted temperature evolution. The DNS and model

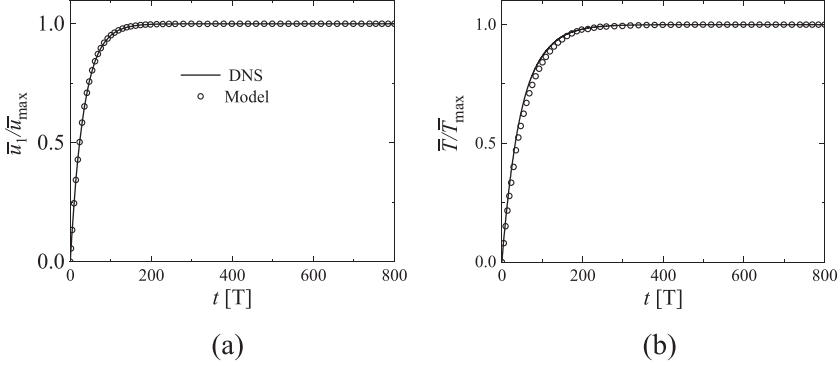


FIG. 11. The changes of coupled velocity and temperature versus time, $Re_l = 1.18 \times 10^4$: (a) velocity and (b) temperature.

results are compared in Fig. 11, which demonstrates that the results of the coupling model for temperature are consistent with those obtained from DNS, despite a minor discrepancy that may be attributed to the error of h . It is worth mentioning that the model coefficients are calculated only with data in steady or statistically steady flow states. Therefore, such satisfactory accuracy implies that the relaxation timescale for microscopic turbulence to achieve approximate steady state is smaller than the timescale of the transient growth of mean velocity and temperature. It is also important to note that temperature reacts more slowly than velocity, as it requires time to adjust to changes in velocity. In conclusion, these coupled macroscopic equations can be utilized to predict the real-time temperature changes.

V. CONCLUSION

The coupled macroscopic equations governing turbulent flow and heat transfer in porous media are derived based on the pore-scale prevalence hypothesis, which asserts that the characteristic length scale for turbulent flow is determined by the pore size. The macroscopic energy equation incorporates the modeling of the heat transfer coefficient and thermal dispersion. This study identifies two primary scaling laws for the heat transfer coefficient between the fluid and solid phases. In the Darcy regime, characterized by small Reynolds numbers, the scaling law is represented as $h \sim Re_p^2$. Conversely, in the Forchheimer regime, applicable to intermediate and large Reynolds numbers, the scaling law is expressed as $h \sim Re_p^{0.5}$. The transition between the two scaling regimes occurs at approximately the Reynolds number $Re_p = 1$.

The thermal dispersion model is incorporated into the macroscopic energy equation. The influence of the macroscopic temperature gradient on energy transport is approximated using a Laplacian term. A local Reynolds number Re_d which quantifies the strength of local momentum dispersion is introduced to calculate the effective thermal diffusivity, analogous to the momentum-dispersion model. The effective thermal diffusivity approaches molecular thermal diffusivity as porosity approaches 1 and tends toward infinity as porosity approaches zero.

The models for the heat transfer coefficient and thermal dispersion have been effectively validated using typical porous media characterized by either one or two length scales. The macroscopic energy equation is coupled with our previous macroscopic momentum equation to simulate turbulent forced thermal convection within the porous media. Despite minimal differences attributed to the Darcy-Forchheimer term, the results demonstrate that the proposed modeling of heat transfer coefficient and thermal dispersion maintains a high level of accuracy across a diverse range of Darcy numbers, Reynolds numbers Re_s , porosities, and Prandtl numbers.

The coupled macroscopic equations are valuable not only for predicting temperature distributions but also for predicting real-time temperature changes, particularly in the field of thermal prediction.

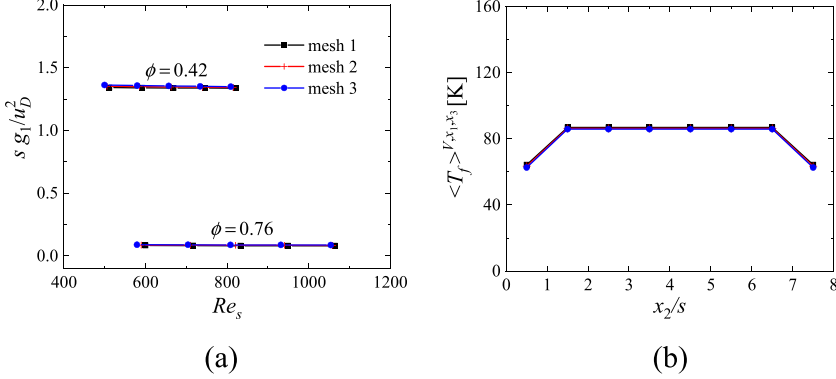


FIG. 12. DNS results of FVM with different mesh resolutions are compared: mesh 1 = 60^3 /REV (solid square, black), mesh 2 = 72^3 /REV (cross, red), mesh 3 = 90^3 /REV (solid circle, blue), turbulent flow. (a) The drag coefficient f_D versus Reynolds number Re_s , $\phi = 0.42$, and $\phi = 0.76$. (b) The REV-averaged temperature $\langle T_f \rangle^{V, x_1, x_3}$ versus the normalized distance from the wall, $Re_s = 717$, $\phi = 0.76$, $Pr = 5$.

The comparison between DNS and model results demonstrates that the macroscopic equations effectively track real-time changes in temperature. The coupled macroscopic equations are particularly beneficial in temperature-sensitive fields.

ACKNOWLEDGMENTS

This research was supported by Zhejiang Provincial Natural Science Foundation of China under Grant No. LQN25E060006 and Zhejiang Normal University under Grant No. YS304024917.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

APPENDIX: VERIFICATION

The finite volume method (FVM) has been extensively applied and validated in our previous research [2,40,44]. In all simulation cases, body-fitted grids are utilized, with the temperature boundary condition of the porous matrix fixed at zero. In this study, we further investigate mesh independence in a porous medium characterized by a single length scale. Figure 12(a) illustrates the relationship between the drag coefficient $f_D = s g_1/u_D^2$ and the Reynolds number Re_s , calculated from various mesh resolutions and a solver utilizing FVM. The porous matrix is composed of cubes with specified parameters $\phi = 0.42$ and $\phi = 0.76$, and three distinct mesh configurations are employed, comprising 60^3 , 72^3 , and 90^3 grid points for each REV. It is evident that the calculated drag coefficient and Reynolds number Re_s exhibit only marginal changes as the number of mesh cells increases. Consequently, it can be concluded that the results demonstrate mesh independence.

To further verify mesh independence, the volume-averaged temperature of three distinct mesh configurations is compared, as illustrated in Fig. 12(b). The Reynolds number Re_s , Prandtl number Pr , and porosity ϕ are 717, 5, and 0.76, respectively. The REV-averaged temperature is calculated for each mesh configuration over time during fully developed turbulent flow. A comparison among the different meshes reveals negligible changes in the REV-averaged temperature. This finding suggests that the results are independent of mesh size.

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